

Dr. Evan E. Seitz

CONTACT INFORMATION

Postdoctoral Researcher
New York, NY 10032 USA

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RESEARCH INTERESTS

Developing interpretable machine learning methods to decode biological complexity — from 3D molecular structure to gene regulation.

CURRENT ACADEMIC APPOINTMENT

Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

Computational Postdoctoral Fellow, March 2022 – present

- Mentors: [Justin Kinney](#) and [Peter Koo](#) (co-P.I.s)
- Affiliations:
[Simons Center for Quantitative Biology](#)
- Summary: My postdoctoral work has produced SEAM and SQUID, two interpretability frameworks for genomic deep learning, including a publication in [Nature Machine Intelligence](#) and an invited talk at [ICLR GEM](#).
- Awards:
 - F32 Individual Postdoctoral Fellowship (2024). Awarded by the National Human Genome Research Institute of the National Institutes of Health. Award Number: F32HG013265.

EDUCATION

Columbia University, New York, NY

Doctor of Philosophy with distinction, May 2017 – January 2022

- Doctoral Thesis: *Analysis of Conformational Continuum and Free-energy Landscapes from Manifold Embedding of Single-particle Cryo-EM Ensembles of Biomolecules*
- Mentor: [Joachim Frank](#)
- Affiliations:
[Department of Biological Sciences](#)
[Department of Biochemistry and Molecular Biophysics](#)
- Summary: My thesis work focused on the development, interpretation and refinement of a geometric machine-learning approach, called ManifoldEM, using manifold embedding to obtain the energy landscape and corresponding continuum of 3D structures of a molecular machine from an ensemble of cryo-EM images afflicted by low signal-to-noise ratio, random rotations and orientations in 3D space, and distortions introduced by microscopy aberrations. A complete description of my accomplishments is available in my thesis [1].
- Awards:
 - John S. Newberry Prize (2022). Awarded to the graduate student in the Department who, in the opinion of the faculty, is the "most promising student of the year in the field of vertebrate zoology". The awardee was chosen by faculty and staff nominations and consideration by the departmental faculty committee for graduate affairs.

Master of Philosophy, May 2017 – October 2020

Master of Arts, May 2017 – May 2019

- Cumulative GPA: 3.74
- Core Courses: *Advanced Genetic Analysis; Cell Biology; Eukaryotic Gene Expression; Genomics of Gene Regulation; Protein Thermodynamics; Structural Biology*
- Elective Courses: *Computational Linear Algebra; Cryo-Electron Microscopy; Statistical Mechanics; Topology*

Pre-thesis Rotations:

- [Califano Lab](#), Fall 2017: I conducted research under the guidance of Dr. Andrea Califano, exploring cell regulatory networks using information-theoretic algorithms (ARACNE, FIRE) to identify a set of maximally-informative DNA sequence motifs associated with the FOXM1 master-regulator pathway implicated in tumorigenesis.
- [Gautier Lab](#), Summer 2017: I conducted research under guidance of Dr. Jean Gautier, investigating the role of genome instability in cancer using various wet lab experiments—such as plasma purification, spectrophotometry, and electrophoresis—to isolate and analyze specific protein-gene interactions responsible for cellular response to DNA damage.

Georgia Institute of Technology, Atlanta, GA

Bachelor of Science in Physics with *Highest Honor*, May 2015 – May 2017

Cumulative GPA: 3.90

- Core Courses: *Classical Mechanics; Differential Equations; Electro and Magnetostatics; Electrodynamics; Linear Algebra for Calculus; Modern Physics; Quantum Mechanics I; Quantum Mechanics II; Statistical Mechanics; Thermodynamics*
- Elective Courses: *Biophysics; Computational Physics; Neurophysics; Nuclei, Particles and Fields; Physics of Living Systems; Probability and Statistics*

Undergraduate Research

- Supervisor: Professor [James Gumbart](#)
- Summary: My undergraduate research in Computational Biophysics, supervised by Dr. James Gumbart, focused on simulating the molecular dynamics of the Light Harvesting Complex-II using NAMD and VMD.

Activities and Awards

- *Sigma Pi Sigma* Physics Honor Society
- Faculty Honors

Georgia State University, Atlanta, GA

Initiated my pursuit towards a second degree (in Physics), May 2014 – May 2015. Transferred from Georgia State University (without degree) to the Georgia Institute of Technology

Cumulative GPA: 4.00

- Core Courses: *Calculus of One Variable I; Calculus of One Variable II; Multivariate Calculus; Principles of Chemistry I; Principles of Chemistry II; Principles of Physics I; Principles of Physics II*
- Elective Courses: *Computer Programming in Python*

Activities and Awards

- President's List

Georgia College, Milledgeville, GA

Bachelor of Arts in Mass Communication, May 2005 – May 2009

Activities and Awards

- *Lambda Pi Eta* Honor Society
- National Chair of International Business Club

REFEREED
PUBLICATIONS

- [1] E. Seitz, D. McCandlish, J. Kinney and P. Koo, "Interpreting cis-regulatory mechanisms from genomic deep neural networks using surrogate models." Nat Mach Intell, 2024.
<https://doi.org/10.1038/s42256-024-00851-5>
- [2] E. Seitz, J. Frank and P. Schwander, "Beyond ManifoldEM: Geometric relationships between manifold embeddings of a continuum of 3D molecular structures and their 2D projections." RSC Digital Discovery, vol. 2, no. 3, pp. 702–717, 2023.
<https://doi.org/10.1039/D2DD00128D>
- [3] E. Seitz, F. Acosta-Reyes, S. Maji, P. Schwander and J. Frank, "Recovery of conformational continuum from single-particle cryo-EM images: Optimization of ManifoldEM informed by ground truth." IEEE Trans Comput Imaging, vol. 8, pp. 462-78, 2022.
<https://ieeexplore.ieee.org/document/9773954>
- [4] T. Sztain et al., "A glycan gate controls opening of the SARS-CoV-2 spike protein," Nat Chem, vol. 13, pp. 963–8, 2021.
<https://www.nature.com/articles/s41557-021-00758-3>
- [5] E. Seitz and J. Frank, "POLARIS: Path of least action analysis on energy landscapes," ACS J Chem Inf Model, vol. 60, no. 5, pp. 2581–90, 2020.
<https://pubs.acs.org/doi/10.1021/acs.jcim.9b01108>

CONFERENCE
TALKS

- [6] E. Seitz, D. McCandlish, J. Kinney and P. Koo. "Mapping the Mechanistic Impact of Genetic Variation with Interpretable Deep Learning". *Keystone Symposia: AI in Molecular Biology*, September 2025.
- [7] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Spotlight Poster presentation: "Decoding the Mechanistic Impact of Genetic Variation on Regulatory Sequences with Deep Learning". *Machine Learning in Computational Biology*, September 2025.
- [8] E. Seitz, D. McCandlish, J. Kinney and P. Koo. "Decoding the Mechanistic Impact of Genetic Variation on Regulatory Sequences with Deep Learning," *Generative and Experimental Perspectives for Biomolecular Design (GEM) Workshop, International Conference on Learning Representations (ICLR)*, Singapore, April 2025.
• URL: <https://openreview.net/forum?id=PtjMeyHcTt>

CONFERENCE
POSTERS

- [9] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Mapping the Mechanistic Impact of Genetic Variation with Interpretable Deep Learning. *Keystone Symposia: AI in Molecular Biology*, September 2025.
- [10] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Decoding the Mechanistic Impact of Genetic Variation on Regulatory Sequences with Deep Learning. *Machine Learning in Computational Biology*, September 2025.
- [11] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Mapping the Mechanistic Impact of Genetic Variation with Interpretable Deep Learning. *Human Genetics and Genomics*, Gordon Research Conference and Gordon Research Seminar, July 2025.
- [12] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Mapping the Mechanistic Impact of Genetic Variation with Interpretable Deep Learning. *Human Genetics and Genomics*, Gordon Research Conference and Gordon Research Seminar, July 2025.

- [13] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Decoding the Mechanistic Impact of Genetic Variation on Regulatory Sequences with Deep Learning. *Biology of Genomes*, Cold Spring Harbor Laboratory, May 2025.
- [14] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Deciphering the determinants of mechanistic variation in regulatory sequences. *Probabilistic Modeling in Genomics*, Cold Spring Harbor Laboratory, March 2025.
- [15] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Deciphering the determinants of mechanistic variation in regulatory sequences. 2025 In-House Symposium, Cold Spring Harbor Laboratory, January 2025.
- [16] E. Seitz, D. McCandlish, J. Kinney and P. Koo. Deciphering the determinants of mechanistic variation in regulatory sequences. *Biological Data Science*, Cold Spring Harbor Laboratory, November 2024.
- [17] E. Seitz, D. McCandlish, J. Kinney and P. Koo. A surrogate modeling framework for interpreting deep neural networks in functional genomics. *Genome Informatics*, Cold Spring Harbor Laboratory, December 2023.
- [18] E. Seitz, J. Kinney and P. Koo. A surrogate modeling framework for interpreting deep neural networks in functional genomics. *The Biology of Genomes*, Cold Spring Harbor Laboratory, May 2023.
- BOOK CHAPTERS
- [19] E. Seitz, J. Frank, POLARIS: Path of Least Action Analysis on Energy Landscapes. In: J. Frank, *Novel Developments in Cryo-EM of Biological Molecules: Resolution in Time and State Space*, Jenny Stanford Publishing, ch. 8, pp. 151–175, 2023. ISBN: 9781003456100.
- [20] T. Sztain, S. Ahn, A. Bogetti, L. Casalino, J. Goldsmith, E. Seitz, R. McCool, F. Kearns, F. Acosta-Reyes, S. Maji, G. Mashayekhi, J. McCammon, A. Ourmazd, J. Frank, J. McLellan, L. Chong, R. Amaro, A Glycan Gate Controls Opening of the SARS-CoV-2 Spike Protein. In: J. Frank, *Novel Developments in Cryo-EM of Biological Molecules: Resolution in Time and State Space*, Jenny Stanford Publishing, ch. 11, pp. 241–256, 2023. ISBN: 9781003456100.
- [21] E. Seitz, F. Acosta-Reyes, S. Maji, P. Schwander, J. Frank, Recovery of Conformational Continuum from Single-Particle Cryo-EM Images: Optimization of ManifoldEM Informed by Ground Truth. In: J. Frank, *Novel Developments in Cryo-EM of Biological Molecules: Resolution in Time and State Space*, Jenny Stanford Publishing, ch. 12, pp. 242–288, 2023. ISBN: 9781003456100.
- OTHER PUBLICATIONS
- [22] E. Seitz. *Analysis of Conformational Continuum and Free-energy Landscapes from Manifold Embedding of Single-particle Cryo-EM Ensembles of Biomolecules*. PhD thesis. Columbia Libraries Academic Commons, 2022.
<https://doi.org/10.7916/4n0v-wa24>
- GRANTS
- Awarded**
- [1] F32 Individual Postdoctoral Fellowship, Awarded by the National Human Genome Research Institute of the National Institutes of Health. Award Number: F32HG013265. June 1, 2024 to May 31, 2027.

ADVISING AND
MENTORING

Undergraduate Research

• **Nika Chuzhoy**

Undergraduate student in Computer Science, California Institute of Technology. Computational research on surrogate modeling for interpreting genomic DNNs (2023)

High School Research

• **Tina Yu**

High school student attending Syosset High School, NY. Computational research on surrogate modeling for interpreting genomic DNNs (2024)

• **Sophia Chen**

High school student attending Farragut High School, NY. Computational research on meta-interpretability of genomic DNNs (2025)

TEACHING
EXPERIENCE

Cold Spring Harbor Laboratory, Cold Spring Harbor, NY

Invited Lecturer

March 5, 2025

- “Decoding the Mechanistic Impact of Genetic Variation on Regulatory Sequences with Deep Learning”, Quantitative Biology and Artificial Intelligence In-house Seminar Series

Invited Lecturer

July 19, 2024

- “From Data to Discovery: Navigating Biological Complexity with Interpretable Machine Learning”, Bioinformatics and Computational Neuroscience Lecture Series

Invited Lecturer

March 6, 2024

- “Deciphering the Regulatory Landscape Encoded by Genomic Deep Neural Networks”, Quantitative Biology and Artificial Intelligence In-house Seminar Series

Invited Lecturer

July 21, 2023

- “From Data to Discovery: Navigating Biological Complexity with Interpretable Machine Learning”, Bioinformatics and Computational Neuroscience Lecture Series

Invited Lecturer

April 12, 2023

- “A Surrogate Modeling Framework for Interpreting Deep Neural Networks in Functional Genomics”, Quantitative Biology and Artificial Intelligence In-house Seminar Series

Columbia University, New York, NY

Invited Lecturer

November 9, 2021

- Cryo-EM Microscopy Center (CEMC) 2021 Fall Workshop

Teaching Assistant, RI University

August 2018 – December 2019

- Assistant Instructor for BCHM GU4323: Biophysical Chemistry (Professors: [Dr. John Hunt](#) and [Dr. Art Palmer](#))
 - Fall 2019
 - Responsible for weekly 1-hour recitation lecture, proctoring and grading. This course covered a rigorous introduction to the theory underlying widely used biophysical methods to understand the behavior of molecules and develop related analytical tools, including applications to biomedical research problems.

- Grader for BIOL UN2005 Intro Biology: Biochemistry, Genetics & Molecular Biology
 - Fall 2018
 - Proctored and graded exams.

Georgia Institute of Technology, Atlanta, GA

Creative Director & Animator, R1 University

May 2016 – February 2017

- Assisted the Department of Physics with the design and animation of educational content for Georgia Tech MOOC (massive online open courses) under advisement of **Dr. Michael Schatz**. Topics ranged from *Introductory Mechanics* to *Electromagnetism*. Duties included both scientific and artistic, beginning with the translation of physical concepts and theories into a visual language, and ending in the full production of animated video content for semester-long courses. After initial production, served as project lead on a team of animators for creating an extensive library of related videos for the university.
- Lesson 1 – Introduction to Electric Fields: <https://vimeo.com/237845454>

Georgia State University, Atlanta, GA

Physics Learning Assistant, R1 University

Spring 2015

- Recruited into the Learning Assistant program to aid in the education of students taking *Principles of Physics* under advisement of **Dr. Joshua Von Korff**. Duties included instruction of two one-hour Physics labs each week, attendance of weekly pedagogy lessons taught by Physics professors, and execution of a weekly theoretical practice lecture with graduate students.

Stanford University, Stanford, CA

Creative Director, Animator

2013, 2015

- Directed a team of animators for creating a series of educational videos detailing scientific research done by **Dr. Jennifer Aaker** and colleagues at Stanford on topics including empathy, humor, and purpose. The following videos premiered at the *Future of Story-Telling Summit* in NYC
 - The Happiness Narrative (2015): <https://vimeo.com/210360824>
 - Persuasion and the Power of Story (2013): <https://vimeo.com/74576399>

**PROFESSIONAL
SERVICE**

Referee Service

- *NeurIPS SPIGM* (2025)
- *NeurIPS AI4Science* (2025)
- *ACM Bioinformatics, Computational Biology, and Health Informatics* (2025)
- *ICML 2025 GenBio Workshop* (2025)
- *RECOMB* (2024)
- *Nature Genetics* (2024)
- *Nature Genetics* (2024)
- *Nature* (2024)
- *Nature* (2024)
- *The American Journal of Human Genetics* (2024)
- *ICML AI4Science Workshop* (2024)
- *Cold Spring Harbor Perspectives in Biology* (2023)
- *NeurIPS AI4Science Workshop* (2023)
- *NeurIPS Generative AI and Biology Workshop* (2023)
- *Nature* (2023)
- *ICML Workshop on Computational Biology* (2023)
- *Genome Research* (2023)
- *Nature Machine Intelligence* (2023)
- *Journal of Chemical Information and Modeling* (2020)

Scientific Illustrations

- Illustrated journal cover for *JCIM: Special Issue on Frontiers in Cryo-EM Modeling*, Volume 60, Issue 5, May 2020: <https://pubs.acs.org/toc/jcisd8/60/5>
- Illustrated cover for *Simons Center for Quantitative Biology: 2024 Annual Report*, December 2024: https://www.cshl.edu/wp-content/uploads/2025/01/2024_SCQB_Annual_Report.pdf

CREATIVE BACKGROUND

Prior to entering science, I worked professionally in 2D/3D animation and motion design. Projects included clients such as Coca-Cola, Cirque du Soleil, and Stanford University. A portfolio and full history are available on my [LinkedIn profile](#).

ADDITIONAL TRAINING

Invited Participant

- CIFAR Deep Learning + Reinforcement Learning Summer School (July 25–29, 2022)
- CSHL Computational Genomics course (Nov. 29–Dec. 6, 2023)

General Participant

- CSHL Mentor Training Workshop based on the curriculum developed by [CIMER](#) (June 2023)
- SUNY Old Westbury Undergraduate Teaching and Pedagogy Workshop on course design, active learning, and inclusive teaching (Fall 2023)

SOFTWARE SKILLS

Computer Programming:

- UNIX shell scripting
- Expertise in Python, with significant experience using TensorFlow, PyTorch, NumPy, Scipy, Pandas, Matplotlib, Mayavi, PyQt, TraitsUI and Anaconda, among others
- Experience in Matlab, R

Scientific Software:

- UCSF Chimera
- PyMOL
- Visual Molecular Dynamics
- RELION
- Phenix
- Prism
- Mathematica

Visualization Software:

- Adobe Illustrator, Photoshop, After Effects (2/2.5D illustration and animation)
- Cinema4D (3D illustration and animation)
- Final Cut Pro (Video editing)

Productivity Software:

- $\text{\LaTeX}$ ($\text{\LaTeX}$, $\text{\BIBTeX}$),
- Microsoft Office, Excel, and PowerPoint
- GitHub

Operating Systems:

- Linux
- Apple OS X

SOFTWARE DISTRIBUTIONS

- E. Seitz et al., "SEAM Python repository: Systematic explanation of attribution-based mechanisms for regulatory genomics," 2024-2025. <https://github.com/evanseitz/seam-nn>
– Read the Docs: <https://seam-nn.readthedocs.io>

- E. Seitz et al., "SQUID Python repository: Interpreting sequence-based deep learning models for regulatory genomics," Zenodo, 2024.
doi:10.5281/zenodo.11060672, <https://github.com/evanseitz/squid-nn>
– Read the Docs: <https://squid-nn.readthedocs.io>
- E. Seitz et al., "ManifoldEM Python repository," Zenodo, 2021.
doi: 10.5281/zenodo.5578874, https://github.com/evanseitz/ManifoldEM_Python
– Video Demonstration:
https://www.dropbox.com/s/pe106oizw4p7uyb/GUI_Overview_VATPase.mp4?dl=0

REFERENCES
AVAILABLE TO
CONTACT

Dr. Peter Koo (e-mail: koo@cshl.edu)

- *I am currently working in Dr. Koo's lab as a Computational Postdoctoral Fellow*
- ◇ Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY 11724 USA

Dr. Justin Kinney (e-mail: jkinnay@cshl.edu)

- *I am currently working in Dr. Kinney's lab as a Computational Postdoctoral Fellow*
- ◇ Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY 11724 USA

Dr. David McCandlish (e-mail: mccandlish@cshl.edu)

- *I am currently collaborating with Dr. McCandlish during my postdoctoral fellowship*
- ◇ Simons Center for Quantitative Biology, Cold Spring Harbor Laboratory, Cold Spring Harbor, NY 11724 USA

Dr. Joachim Frank (e-mail: jf2192@cumc.columbia.edu; phone: +1-646-770-4527)

- *Dr. Frank was my Ph.D. supervisor*
- ◇ Department of Biochemistry and Molecular Biophysics, Columbia University Medical Center, New York, NY 10032 USA
- ★ Department of Biological Sciences, Columbia University, New York, NY 10027 USA

Dr. Peter Schwander (e-mail: pschwan@uwm.edu)

- *Dr. Schwander was a member of my Ph.D. dissertation defense committee*
- ◇ Department of Physics, University of Wisconsin-Milwaukee, Milwaukee, WI 53211 USA

Dr. Rommie Amaro (e-mail: ramaro@ucsd.edu)

- *Dr. Amaro was a member of my Ph.D. dissertation defense committee*
- ◇ Department of Chemistry and Biochemistry, University of California-San Diego, La Jolla, CA 92093 USA

Dr. John Hunt (e-mail: jfh21@columbia.edu)

- *Dr. Hunt was a member of my Ph.D. dissertation defense committee*
- ◇ Department of Biological Sciences, Columbia University, New York, NY 10027 USA

Dr. Liang Tong (e-mail: ltong@columbia.edu)

- *Dr. Tong was a member of my Ph.D. dissertation defense committee*
- ◇ Department of Biological Sciences, Columbia University, New York, NY 10027 USA

Additional references available upon request.